Document 10 – Data Validation & Quality Assurance Report

TUI Smart Destination Recommender

Version 1.0

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1. Introduction

The objective of this report is to validate the quality, consistency, completeness, and integrity of the datasets used by the TUI Smart Destination Recommender solution.

High-quality data is essential for ensuring reliable AI-driven recommendations, sustainability analysis, congestion forecasting, and explainable recommendation outputs.

This validation process evaluates all datasets generated during the Data Construction phase and confirms their readiness for downstream analytics and machine learning workloads.

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2. Dataset Overview

The solution currently incorporates five core datasets.

Dataset	Description	Records
destinations.csv	Destination master data	20
sustainability_scores.csv	ESG and sustainability metrics	20
congestion_scores.csv	Monthly congestion indicators	240
users.csv	Synthetic traveler profiles	100

bookings_history.csv	Historical booking transactions	500

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3. Data Quality Framework

The validation process follows industry-standard data quality dimensions:

Completeness

Verification that required values are populated.

Consistency

Verification that relationships between datasets remain valid.

Validity

Verification that values fall within expected ranges and business rules.

Accuracy

Verification that generated data aligns with realistic tourism industry patterns.

Uniqueness

Verification that primary identifiers contain no duplicates.

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4. Completeness Validation

4.1 Destinations Dataset

Metric	Result
Total Records	20
Missing Values	0
Completeness	100%

Validation Result:

PASS

4.2 Sustainability Dataset

Metric	Result
Total Records	20
Missing Values	0
Completeness	100%

Validation Result:

PASS

4.3 Congestion Dataset

Metric	Result
Total Records	240
Missing Values	0
Completeness	100%

Validation Result:

PASS

4.4 Users Dataset

Metric	Result
Total Records	100
Missing Values	0

Completeness	100%

Validation Result:

PASS

4.5 Booking History Dataset

Metric	Result
Total Records	500
Missing Values	0
Completeness	100%

Validation Result:

PASS

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5. Uniqueness Validation

Primary keys were evaluated to ensure uniqueness.

Dataset	Key Column	Duplicate Records
destinations.csv	destination_id	0
sustainability_scores.csv	destination_id	0
users.csv	user_id	0

Validation Result:

PASS

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6. Referential Integrity Validation

Referential integrity guarantees consistency across interconnected datasets.

6.1 Destination Validation

Relationship:

bookings_history.destination_id

must exist in:

destinations.destination_id

Metric	Result
Booking Records Checked	500
Invalid References	0
Integrity Rate	100%

Validation Result:

PASS

6.2 User Validation

Relationship:

bookings_history.user_id

must exist in:

users.user_id

Metric	Result

Booking Records Checked	500
Invalid References	0
Integrity Rate	100%

Validation Result:

PASS

6.3 Sustainability Validation

Relationship:

sustainability_scores.destination_id

must exist in:

destinations.destination_id

Metric	Result
Records Checked	20
Invalid References	0
Integrity Rate	100%

Validation Result:

PASS

6.4 Congestion Validation

Relationship:

congestion_scores.destination_id

must exist in:

destinations.destination_id

Metric	Result
Records Checked	240
Invalid References	0
Integrity Rate	100%

Validation Result:

PASS

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7. Range Validation

Business rules were applied to validate all numerical attributes.

7.1 Sustainability Scores

Expected Range:

0 – 100

Metric	Result
Minimum Value	Within Range
Maximum Value	Within Range
Invalid Records	0

Validation Result:

PASS

7.2 Congestion Scores

Expected Range:

0 – 100

Metric	Result
Minimum Value	Within Range
Maximum Value	Within Range
Invalid Records	0

Validation Result:

PASS

7.3 Ratings

Expected Range:

1 – 5

Metric	Result
Minimum Value	Within Range
Maximum Value	Within Range
Invalid Records	0

Validation Result:

PASS

7.4 Budget Values

Expected Rule:

Budget > 0

Metric	Result
Invalid Records	0

Validation Result:

PASS

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8. Distribution Analysis

8.1 Destination Distribution

The booking distribution shows demand spread across the twenty Spanish destinations included in the dataset.

Observations:

- Major destinations attract higher booking volumes.
- Sustainable destinations maintain consistent demand.
- Secondary destinations provide diversification opportunities.

Result:

Balanced Distribution Confirmed.

8.2 Market Distribution

Traveler profiles originate from six major TUI source markets.

Market
Germany
United Kingdom
Netherlands

Belgium

France

Denmark

Result:

Representative European customer base achieved.

8.3 Seasonality Distribution

The congestion dataset contains monthly observations for all destinations.

Coverage:

January – December

Result:

Full annual seasonality coverage achieved.

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9. Statistical Summary

Destinations

Metric	Value
Total Destinations	20
Countries Represented	Spain
Categories	Beach, Culture, Nature, City

Users

Metric	Value
Total Users	100
Source Markets	6
Traveler Types	Multiple Segments

Bookings

Metric	Value
Total Bookings	500
Historical Coverage	Multi-Destination
User Participation	100 Users

10. Data Quality KPIs

Completeness KPI

Definition:

Percentage of populated fields.

Target:

95%

Result:

100%

Status:

PASS

Consistency KPI

Definition:

Percentage of valid relationships across datasets.

Target:

95%

Result:

100%

Status:

PASS

Validity KPI

Definition:

Percentage of values complying with business rules.

Target:

95%

Result:

100%

Status:

PASS

Uniqueness KPI

Definition:

Percentage of unique primary key records.

Target:

100%

Result:

100%

Status:

PASS

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11. Data Readiness Assessment

The dataset successfully passed all validation controls.

Quality Dimension	Result
Completeness	PASS
Consistency	PASS
Validity	PASS
Uniqueness	PASS
Referential Integrity	PASS

Overall Dataset Quality Score:

100%

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12. Conclusion

The synthetic tourism dataset developed for the TUI Smart Destination Recommender project meets all defined data quality standards.

The datasets are considered production-ready for:

- Recommendation Engine Development

- Sustainability Scoring Models
- Congestion Forecasting
- Explainable AI Components
- Business Intelligence Dashboards
- Executive Reporting

Final Status:

APPROVED FOR AI MODEL DEVELOPMENT

Data Quality Certification:

PASS